SCALING UP TRAINING CONTINUOUS LATENT REASONING: ON THE POWER OF SOFT THINKING AND REINFORCEMENT LEARNING

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ABSTRACT

We proposed a novel paradigm for empowering reasoning LLMs by combining Soft Thinking, a variant of continuous reasoning, with Large Scale Reinforcement Learning. Unlike previous methods of with-training continuous reasoning, our approach directly demonstrates State-of-the-Art performance on difficult reasoning-required benchmarks such as AIME and MATH-500, showcasing significant stability and full potential to scaling up.

1 INTRODUCTION

The Chain of Continuous Thought (COCONUT) (Hao et al., 2024) explores the potential of the latent layer of reasoning as an alternative to the traditional explicit CoT (Wei et al., 2022) in both the reasoning procedure and the response procedure. Its previous work has shown that the latent layer of continuous reasoning can reduce unnecessary statements and focus on the core of the problem.

These models are issued from general-purpose language, trained in supervised steps of a reasoning procedure, and progressively increase their layers of latent reasoning. It is impractical due to the lack of customized reasoning data, and progressive training is significantly less convenient than explicit CoT. Hence, we provide a new approach to implementing this latent reasoning layer, fine-tuning from explicit reasoning models.

In January, the DeepSeek team proposed a distillation-based method to implement a reasoning model through reinforcement learning (DeepSeek-AI, 2025), and applied the approach to Qwen-2.5B (Yang et al., 2024b) to a reasoning model. Given the limitations of a 2.5B parameter model in reasoning tasks, we aim to implement COCONUT on the small model and find its potential in reasoning compared to the original explicit reasoning model.

2 RELATED WORK

Explicit Reasoning Models and Large-scale Reinforcement Learning Advances in explicit reasoning models, including OpenAI o1 & o3 (OpenAI, 2025; 2024) and DeepSeek R1 (DeepSeek-AI, 2025), demonstrate their strong ability in mathematical reasoning. Based on Chain-of-Thought (Wei et al., 2022), just through prompts, models achieved significant improvements on arithmetic, commonsense, and symbolic reasoning tasks with *explicitly generate* intermediate reasoning text.

Several novel paradigms were also created from explicit reasoning models (Wang et al., 2023; Yao et al., 2023; Besta et al., 2024). In this way, explicit reasoning creates a new level of mathematical reasoning tasks.

Ouyang et al. (2022) showed how GPT-3 can be fine-tuned with reinforcement learning via Proximal Policy Optimization (PPO) (Schulman et al., 2017), a reinforcement learning algorithm that optimizes the agent's policy by maximizing the expected reward while ensuring that the new policy does not deviate too much from the old policy. Shao et al. (2024) proposed Group Relative Policy Optimization (GRPO), which is a new reinforcement learning algorithm that overcame the limitations of PPO, including the need for a large number of samples and the difficulty of tuning hyperparameters. Yu et al. (2025) demonstrated the effectiveness of Decoupled Clip and Dynamic

Sampling Policy Optimization (DAPO) in large-scale reinforcement learning, which performed best in several benchmarks on Qwen2.5-32B Yang et al. (2024b) solely based on pure reinforcement learning and the base instruction model of Qwen.

Chain of Continuous Thought Previous works have already found that LLMs may involve hidden computation steps, which are not explicitly generated in text (Yang et al., 2024c; Chen et al., 2023). Driven by this spirit, Hao et al. (2024) proposed COCONUT, assigning models to "reason in a latent continuous space" through specifically designed training objectives, and demonstrated its potential compared to explicit Chain-of-Thought reasoning procedure, especially on tasks requiring backtracking. Geiping et al. (2025) introduced a new recurrent transformer architecture that lets models perform multiple hidden reasoning iterations per token. A recurrent block can be looped arbitrarily at inference to refine the hidden state before producing the next token. Chen et al. (2025) argued that LLMs may involve a born ability to reason internally, suggesting that LLMs can internally learn better reasoning policies in the latent space when given appropriate training signals. Saunshi et al. (2025) demonstrated that the critical impact of model capabilities lies in its depth instead of parameter size. In this way, they proposed *Looped Transformer*, which generated latent thoughts and simulated Chain-of-Thought reasoning procedures internally.

Compared to explicit reasoning, latent reasoning paradigms leverage latent spaces, promoting efficiency and accuracy. However, they still face problems with scaling up and stability.

3 METHODOLOGY

In this section, describe our training paradigm that leverages the Reinforcement Learning (RL) paradigm to solidify the *Soft Thinking* (Zhang et al., 2025) paradigm, which uses concept tokens to represent the latent reasoning steps of the model, retaining the full probabilities of tokens rather than decoding to a discrete one.

3.1 MODIFIED SOFT THINKING

Preliminaries. We briefly review core concepts of Soft Thinking proposed by Zhang et al., which serves the foundation of our work.

Let $ct := p \in \Delta^{|V|-1}$ denotes the "probability distribution over the vocabulary" at any intermediate step (i.e., reasoning step) of the language model, which may allow the model to explore different paths simultaneously. By convexly combining all embedding vectors, as well as concept tokens here, we can obtain the "continuous concept space," as denoted in 1

$$\mathcal{C} \;=\; \left\{ \sum_{k=1}^{|V|} \alpha_k \, e(k) \;:\; \alpha \in \Delta^{|V|-1} \right\} \subset \mathbb{R}^d, \tag{1}$$

At each step, the model can produce a concept token rather than explicit decoded token, and we can inject the concept token back into the model, as denoted in 2.

$$\tilde{e}_{\text{next}} = \sum_{k=1}^{|V|} ct[k] \, e(k) = \sum_{k=1}^{|V|} p[k] \, e(k) \in \mathcal{C}. \tag{2}$$

Refinement. We fortuitously found that in the process of computing concept probabilities, changing the order of applying Softmax and then gather the logits and apply Softmax, which can slightly improve the model's performance.

By applying Softmax later, the numerical stability holds better and reduces extreme parameters, which can eventually result to better and more stable reasoning procedures.

We load our empirical data in the zero-training stage (i.e., the initial rollout) in . . . (experiments section not conducted yet.)

3.2 Reinforcement Learning

Algorithm. We adopt the Clipped IS-weight Policy Optimization (CISPO) (MiniMax et al., 2025) algorithm in our pipeline, which can be denoted in 3.

$$\mathcal{J}_{\text{CISPO}}(\theta) = \mathbb{E}_{(q,a)\sim\mathcal{D},\{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^{G}|o_i|} \sum_{i=1}^{G}\sum_{t=1}^{|o_i|} \texttt{sg}(\hat{r}_{i,t}(\theta))\hat{A}_{i,t}\log\pi_\theta(o_{i,t} \mid q, o_{i,<t}) \right], \tag{3}$$

where $\texttt{sg}(\cdot)$ denotes the stop-gradient operation, and $\hat{r}_{i,t}(\theta)$ is the clipped IS weight:

$$\hat{r}_{i,t}(\theta) = \text{clip}\left(r_{i,t}(\theta), 1-\epsilon_{low}^{IS}, 1+\epsilon_{high}^{IS}\right).$$

Reward. We verify answers (Kydlíček) that is wrapped with `\boxed` block, and set the reward as follows:

$$r = \begin{cases} 2, & \text{if correct} \\ -1, & \text{if incorrect} \end{cases}$$

3.2.1 Swapping

In the rollout stage of the GRPO (Shao et al., 2024) algorithm and its followers such as DAPO (Yu et al., 2025) and CISPO (MiniMax et al., 2025), to each prompt the algorithms require sampling multiple answers, i.e., doing group sampling, which is a crucial factor of more accurate advantage estimation. Latest value-model-based algorithms, including PPO (Schulman et al., 2017) leveraged in Open-Reasoner-Zero (Hu et al., 2025) and VAPO (Yue et al., 2025), recommend applying group sampling for performance enhancement.

However, just as the Greedy Sampling, the original Soft Thinking sampling holds no diversity, which means that multiple sampling will only gain one unique answer, thus making group sampling impossible.

To unlock the power of group sampling, we propose a "swapping" technique to add diversity to the sampling process. Specifically, for a output logits tensor $\boldsymbol{l_t}$ on the t-th token, perform the following computation:

$$\begin{aligned} \boldsymbol{p} &= \text{Softmax}(\boldsymbol{l}_{t+1}/T, \text{dim} = -1) \\ \text{topk\_probs}, \text{topk\_indices} &= \text{Topk}(\boldsymbol{p}, k = 10) \\ \text{token}_{t+1} &= \text{MultiNomial}(\text{topk\_probs}) \end{aligned}$$

3.3 Self-Distillation

4 Experiments

4.1 Setup

We use Qwen2.5-1.7B (Yang et al., 2024a;b) as our base model, which is a 1.5B-parameter model with 32 layers. Since Qwen3-1.7B can be altered in both reasoning mode and non-reasoning mode, we conduct the reinforcement learning training on the non-reasoning mode, demonstrating the feasibility of our methodology on general-purpose models, and compare the result with the reasoning mode. Except for the reasoning mode itself, we also compare the performance with the distillation method from DeepSeek R1 (DeepSeek-AI, 2025).

We trained our model with Low-Rank Adaption (LoRA) (Hu et al., 2022) adaption to reduce the training overhead, and successfully trained the model on single GeForce RTX 4090 with 24GB memory. We adapted PiSSA (Meng et al., 2024) with 32 iterations in initialization, and set the rank 16, LoRA alpha 32, and LoRA dropout 0.01.

4.2 DATASETS

We mixed the training dataset with OpenR1 Math 220k (Hugging Face, 2025), DAPO Math 17k (Yu et al., 2025), and MATH (Hendrycks et al., 2021). We preprocessed those datasets to remove the original reasoning text, leaving only the question and answer pairs. The training dataset contains roughly 1.88 million samples, and we randomly select about 100 questions in each sample.

5 DISCUSSION

6 CONCLUSION

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A APPENDIX